NXT WAVE

WELCOME

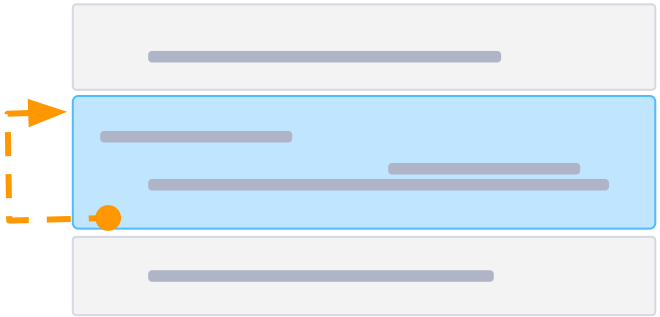
CCBP 4.0

21st Century Skills

For Loop

Loops

Loops allow us to **execute**
a block of code **several times**



Loops

In Python, there are two primary ways for looping



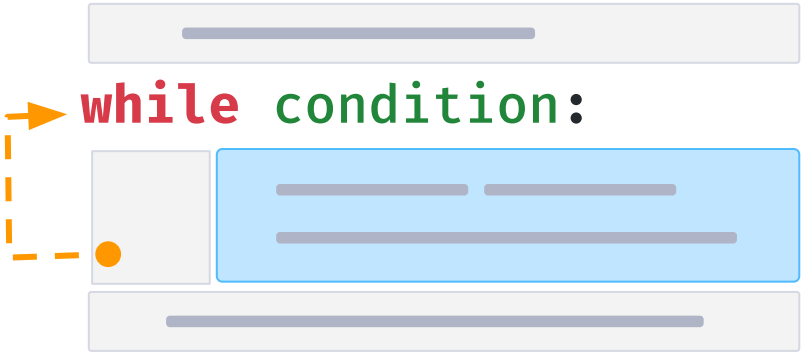
`While Loop`



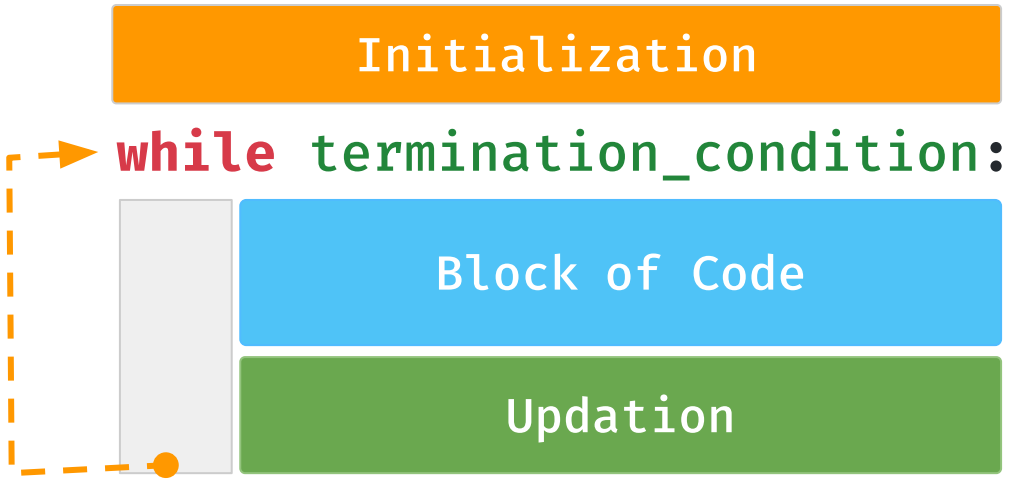
`For Loop`

While Loop

To execute a block of code several times as long as the condition is True



While Loop


Initialization
while termination_condition:
Block of Code
Updation

While Loop

Code

```
a = int(input())
counter = 0
while counter < 3:
    a = a + 1
    print(a)
    counter = counter + 1
print("End")
```

Initialization

Termination Condition

Updation

Daily Challenge

Identify The Mistake

Code

```
a = input()
counter = 0
length_of_a = len(a)
while counter < (length_of_a - 1):
    print(a[counter])
    counter = counter + 1
```

Input

```
Python
```

Expected Output

```
P
y
t
h
o
n
```

Daily Challenge

Possible Solutions

```
a = input()
counter = 0
length_of_a = len(a)
while counter < length_of_a:
    print(a[counter])
    counter = counter + 1
```

Code

```
a = input()
counter = 0
length_of_a = len(a)
while counter <= (length_of_a-1):
    print(a[counter])
    counter = counter + 1
```

Daily Challenge

Possible Solutions

Code

```
a = input()
counter = -1
length_of_a = len(a)
while counter < (length_of_a-1):
    print(a[counter+1])
    counter = counter + 1
```

Daily Challenge

Identify The Mistake

```
a = input()
counter = 0
length_of_a = len(a)
while counter < length_of_a:
    print(a[counter])
    counter = counter + 1
```

Input

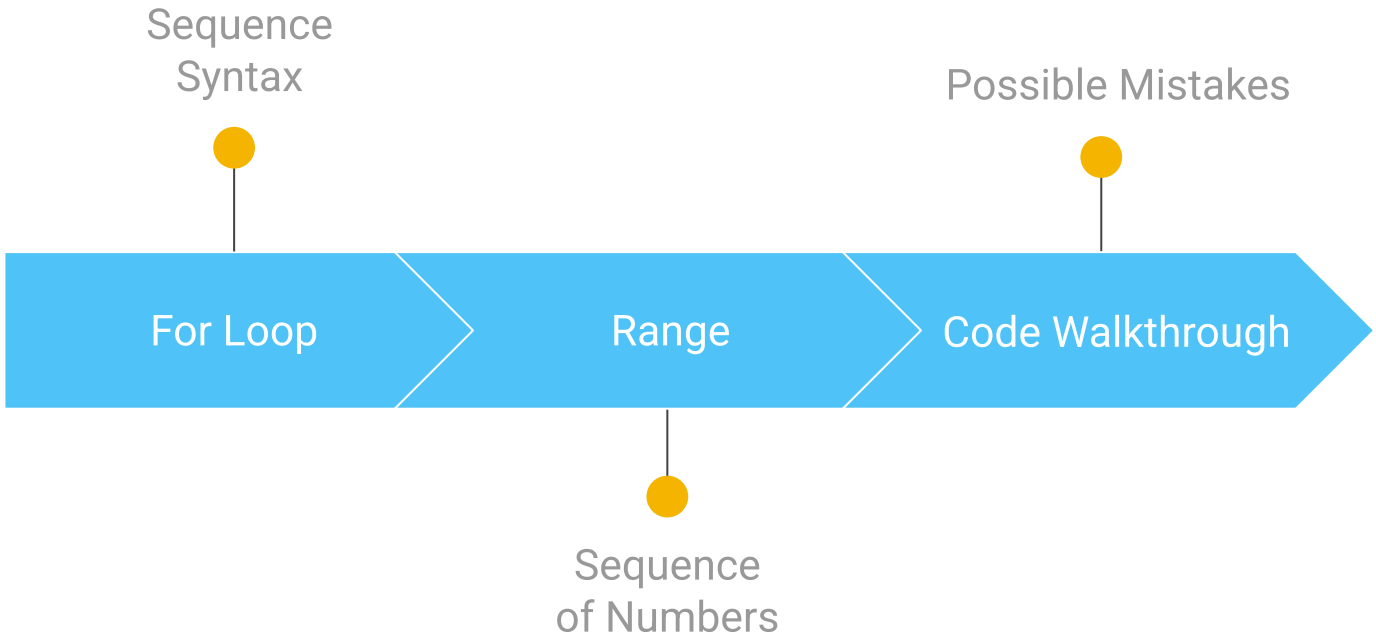
```
Python
```

Output

```
P
y
t
h
o
n
```

Agenda for Today's Session

For Loop



Sequence of Characters

"Python"



A String is a

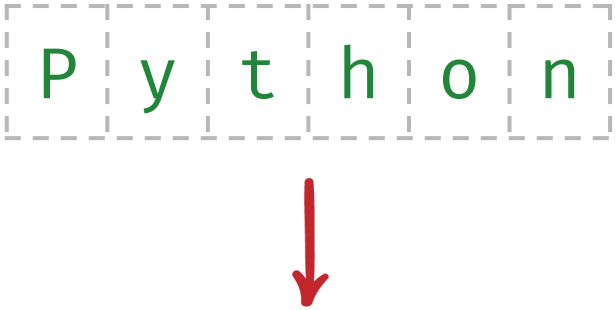
Sequence of Characters

P	y	t	h	o	n



Individual Characters

Sequence of Characters



A block of code
should be executed
for each character of string

Can we simplify this?

Code

```
a = "Python"
length_of_a = len(a)
counter = 0
while counter < length_of_a:
    print(a[counter])
    counter = counter + 1
```

Initialization

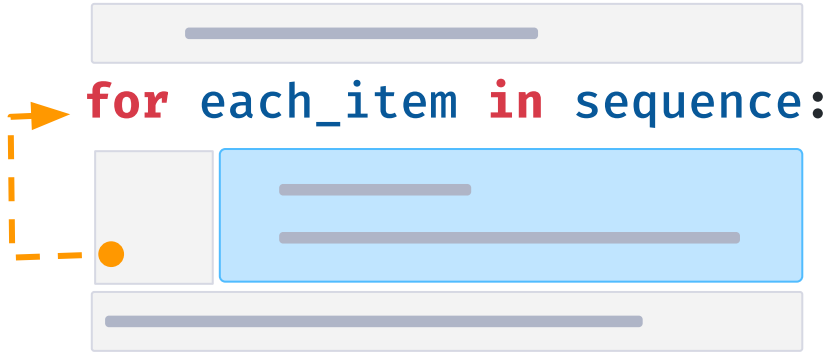
Termination Condition

Updation

For Loop

for statement iterates over each item of a sequence

- Sequence of Characters (string)
- Sequence of numbers, etc.



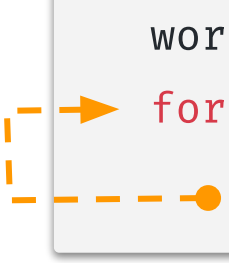
Iterate Over Characters

For Loop

```
for each_item in sequence:
    Repeating
    Block of Code
```

Code

```
word = "Python"
for each_char in word:
    print(each_char)
```



Iterate Over Characters

For Loop

Code

```
word = "Python"
for each_char in word:
    print(each_char)
```



Initialization, termination condition and updation are not required

Iterate Over Characters

For Loop

Code

```
word = "Python"
for each_char in word:
    print(each_char)
```

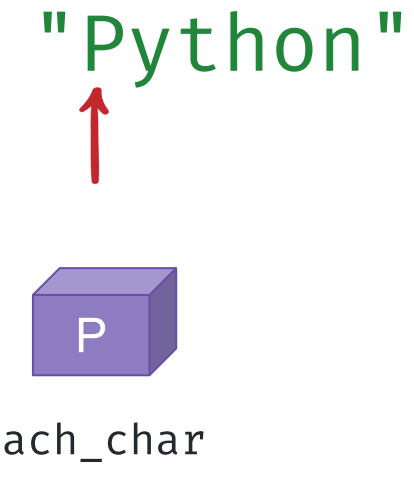
"Python"

Iterate Over Characters

For Loop

Code

```
word = "Python"
for each_char in word:
    print(each_char)
```



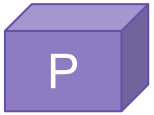
Iterate Over Characters

For Loop

Code

```
word = "Python"
for each_char in word:
    print(each_char)
```





each_char

Output

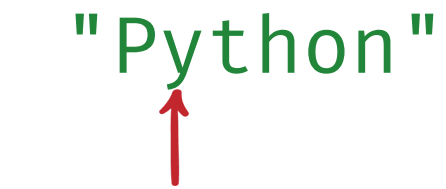
P

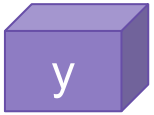
Iterate Over Characters

For Loop

Code

```
word = "Python"
for each_char in word:
    print(each_char)
```





each_char

Output

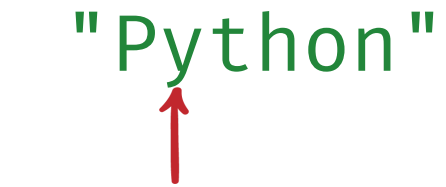
P

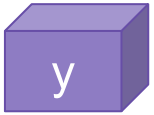
Iterate Over Characters

For Loop

Code

```
word = "Python"
for each_char in word:
    print(each_char)
```





each_char

Output

P

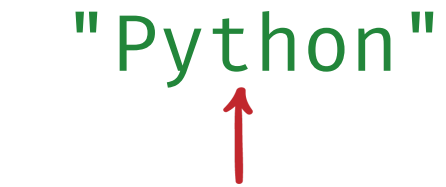
y

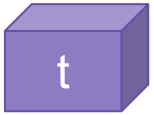
Iterate Over Characters

For Loop

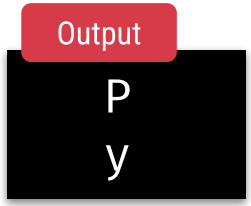
Code

```
word = "Python"
for each_char in word:
    print(each_char)
```





each_char



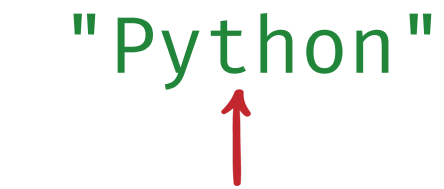
NXT WAVE

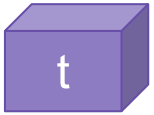
Iterate Over Characters

For Loop

Code

```
word = "Python"
for each_char in word:
    print(each_char)
```





each_char

Output

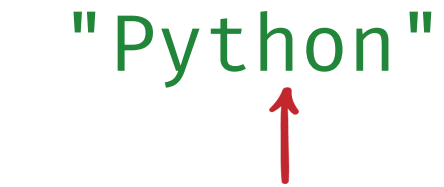
```
P
y
t
```

Iterate Over Characters

For Loop

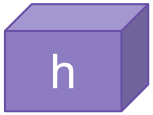
Code

```
word = "Python"
for each_char in word:
    print(each_char)
```



Output

```
P
y
t
```



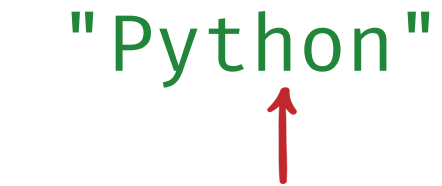
each_char

Iterate Over Characters

For Loop

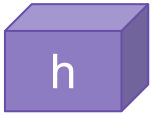
Code

```
word = "Python"
for each_char in word:
    print(each_char)
```



Output

```
P
y
t
h
```



each_char

Iterate Over Characters

For Loop

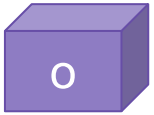
Code

```
word = "Python"
for each_char in word:
    print(each_char)
```



Output

```
P
y
t
h
```



each_char

Iterate Over Characters

For Loop

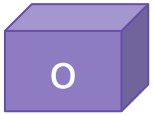
Code

```
word = "Python"
for each_char in word:
    print(each_char)
```



Output

```
P
y
t
h
o
```



each_char

Iterate Over Characters

For Loop

Code

```
word = "Python"
for each_char in word:
    print(each_char)
```



Output

```
P
y
t
h
o
```



each_char

NXT WAVE

Iterate Over Characters

For Loop

Code

```
word = "Python"
for each_char in word:
    print(each_char)
```





each_char

Output

```
P
y
t
h
o
n
```

Iterate Over Characters

For Loop

Code

```
word = "Python"
for each_char in word:
    print(each_char)
```

"Python"

Output

```
P
y
t
h
o
n
```

NXT WAVE

For Loop

Iterate Over Numbers

How to generate
a sequence of numbers?

```
for each_number in sequence_of_numbers:
    Repeating Block of Code
```

Iterate Over Numbers

Range

`range(n)`

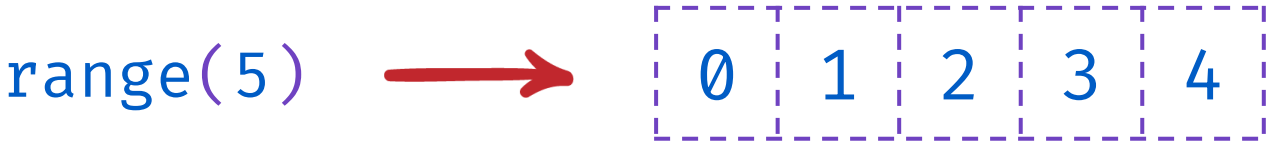


Stops before `n`

(`n` is not included)

Generates a sequence of integers starting from `0`

Sequence of Integers



Gives sequence of integers

Iterate Over Numbers

Range

Code

```
for number in range(3):
    print(number)
```

Output

```
0
1
2
```

Example

Right Angle Triangular Pattern

Write a program to print
right angle triangular pattern,
where integer N is given as input

Input

```
4
```

Output

```
*
* *
* * *
* * * *
```

NXT WAVE

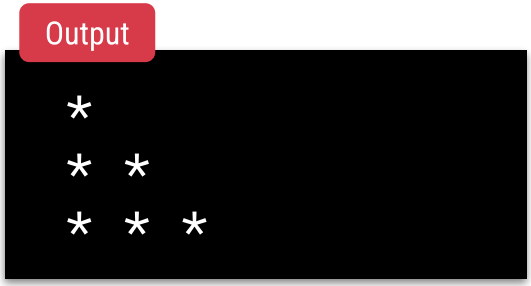
Right Angle Triangular Pattern

Input / Output Format

Input Format

First line will contain a positive integer





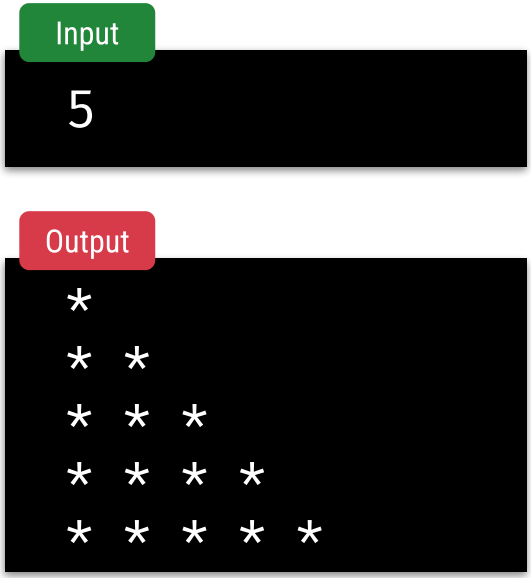
Right Angle Triangular Pattern

Input / Output Format

Output Format

Print right angle triangular pattern

*Note: There is a space after every *****



Iterate Over Numbers

Range

Code

```
n = int(input())
for counter in range(n):
    print("* " * (counter+1))
```

Input

```
4
```

Output

```
*
* *
* * *
* * * *
```

Iterate Over Numbers

Range

Code

```
n = int(input())
for counter in range(n):
    print("* " * (counter+1))
```

Instead of starting from `0`, Can we specify start index of range function ?

Iterate Over Numbers

Range

Generates a sequence of numbers starting from this



```
range(start, end)
```



Stops before **end**

(**end** is not included)

Sequence of Integers



Gives sequence of integers

Iterate Over Numbers

Range

Code

```
for number in range(5, 8):
    print(number)
```

Output

```
5
6
7
```

Iterate Over Numbers

For Loop

```
n = int(input())
for counter in range(1, n+1):
    print("* " * counter)
```

Input

```
4
```

Output

```
*
* *
* * *
* * * *
```

NXT WAVE

Possible Mistakes

Code

```
n = int(input())
for counter in range(1, n):
    print("* " * counter)
```

Input

```
4
```

Output

```
*
* *
* * *
```

Key Takeaways

- Sequence
- For Loop
  - Iterating Over a Sequence
- Range
  - Sequence of Numbers
- Code Walkthrough
- Possible Mistakes



Practice



Feedback



Daily Challenge

Identify The Mistake

Code

```
a = input()
len_of_a = len(a)
b = ""
for i in range(1,len_of_a):
    b = b + "-" + a[i]
print(b)
```

Input

```
Python
```

Expected Output

```
P-y-t-h-o-n
```

Next Session

String Methods

NXT WAVE

THANK YOU

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ALL THE BEST